

Thornado Cash

Nodes

- 67% FROST vault with state machine
- 7-day churning, one in standby (the baddest node)
- Permanent node slots
- Nodes bond $10 + 2.5n$ BTC
- Bond can never leave the system, unless Shutdown
- Nodes can sell node slots in an auction market
- Slot is traded once highest bidder chosen
- Node must churn out, new node churns into that slot
- Nodes can be churned out if offline for an entire churn cycle
- Offline nodes are no observations or TSS Signing for a churn cycle
- Offline nodes pay a 1% penalty (37% loss every year)

Shutdown

- 67% of nodes vote to shutdown
- 30 days of maintenance mode, no further deposits
- All principle returned to nodes
- Fees, Abandoned deposits split equally

Client Mirrors

- All nodes must "vouch" for a client-mirror with SSL
- Client site operates a site that content hashes to a known hash - non-scam site
- If active nodes can validate the site, then nodes can enter the set
- Sites are checked once a week, if down, nodes earn penalty points
- Client also attached to light node, light node broadcasts tx and queries state machine

UX

- Users go to client site, a HD mnemonic is generated
- Users conduct a proof of work to request a deposit address (targeting 30sec on iPhone/browser)
- Users can then deposit any amount of bitcoin
- Once deposit received and confirmed, client can then convert the deposit into a set of notes with a zero knowledge proof, smallest to largest
- 10, 1, 0.1, 0.01 etc
- Each index is an index on the HD wallet path (BIP84)
- ZK Proof is proof that the user is entitled to a withdrawal, but the deposit is unowned.

Deposit -> Split -> Withdraw

ZK

- The deposit address and amount is known.
- The amount is converted to equal-denom notes
- The user proves they deposited an amount, which entitles them to a set of notes
- The user later brings a note, but the note is completely fungible with all other equal denomination notes
- The user places a withdrawal address and submits
- If the proof is valid, the withdrawal submission is gasless (no relayer)

Protection

- Privacy is attained by distance between deposit and withdrawal, controlled by the user
- If user immediately withdraws, no privacy
- Nodes cannot unilaterally move funds, no custodian service
- Nodes are regularly churned, means anyone can leave or reenter
- Permissionless entry-exit
- Funds are not churned for privacy, funds are only churned for the TSS keyset change
- Dev Fund is 5% of Node Slot sales

- Fees are charged only on outbound fees, not against transaction size

Node Income

- Charged as TSS gas handling outbound fees 100k Sats (\$74)
- Rounded up from smallest 0.001 BTC (max \$74)
- Fees accumulate until reach $30 * 0.01$ BTC bucket
- A node operator signs for the bucket, which locks it.
- Each Node Operators sign for a 0.01 BTC note each "I prove that I am a node and I have not yet claimed in this bucket" but not reveal who they are or if they claimed.
- Node Operators can batch sign multiple buckets

Parameters

- Outbound Fee size
- Churn Interval
- Fee Bucket

Network Stack

- Landing Page links out to mirrors, X
- Client-mirrors + light nodes (one click deploy), GitHub registry release hash
- Node + BitcoinD + TSS